William Lane Craig

The Kalam Cosmological Argument



According to the kalam cosmological argument developed by medieval Arabic philosophers, the universe had a cause of its existence because it had a beginning, and whatever has a beginning must have a cause. William Lane Craig (b. 1949) presents four arguments, two from philosophy and two from physics, to support the claim that the universe had a beginning. One philosophical argument shows that an actual infinite cannot exist because it leads to the absurdity that the whole is not greater than its part. The other shows that if an actual infinite could exist, one could not traverse it, which would mean that one could not reach this particular point in time; in an infinite time one would have already reached it. The dual supporting arguments from physics appeal to the Big Bang model of the origin of the universe to confirm a beginning of the universe, and that according to the second law of thermodynamics, if the universe has existed infinitely, by now we should have already been gobbled up by black holes or suffered a "heat death." In the end, Craig concludes that since the universe had a beginning, it was caused, and that the cause had to be personal, not natural.

"The first question which should rightly be asked," wrote G. W. F. Leibniz, is "*Why is there something rather than nothing?*"[1] This question does seem to possess a profound existential force, which has been felt by some of mankind's greatest thinkers. . . . Now there is one form of the cosmological argument that aims precisely at the demonstration that the universe had a beginning in time.[2] Originating in the efforts of Christian theologians to refute the Greek doctrine of the eternity of matter, this argument was developed into sophisticated formulations by medieval Islamic and Jewish theologians, who in turn passed it back to the Latin West. The argument thus has a broad intersectarian appeal, having been defended by Muslims, Jews, and Christians, both Catholic and Protestant. This argument, which I have called the kalam cosmological argument, can be exhibited as follows:

From William Lane Craig, "The Existence of God and the Beginning of the Universe," *Truth: A Journal of Modern Thought* 3 (1991): 85–96 (posted on his website).

(1.) Whatever begins to exist has a cause of its existence.
(2.) The universe began to exist.
  (2.1) Argument based on the impossibility of an actual infinite.
    (2.11) An actual infinite cannot exist.
    (2.12) An infinite temporal regress of events is an actual infinite.
    (2.13) Therefore, an infinite temporal regress of events cannot exist.
  (2.2) Argument based on the impossibility of the formation of an actual infinite by successive addition.
    (2.21) A collection formed by successive addition cannot be actually infinite.
    (2.22) The temporal series of past events is a collection formed by successive addition.
    (2.23) Therefore, the temporal series of past events cannot be actually infinite.
(3.) Therefore, the universe has a cause of its existence.

Let us examine this argument more closely.

Defense of the Kalam Cosmological Argument

Second Premise

Clearly, the crucial premise in this argument is (2), and two independent arguments are offered in support of it. Let us therefore turn first to an examination of the supporting arguments.

First Supporting Argument

In order to understand (2.1), we need to understand the difference between a potential infinite and an actual infinite. Crudely put, a potential infinite is a collection which is increasing toward infinity as a limit but never gets there. Such a collection is really indefinite, not infinite. . . . An actual infinite is a collection in which the number of members really *is* infinite. The collection is not growing toward infinity; it is infinite, it is "complete." . . . Now (2.11) maintains not that a potentially infinite number of things cannot exist, but that an actually infinite number of things cannot exist. For if an actually infinite number of things could exist, this would spawn all sorts of absurdities.

Perhaps the best way to bring home the truth of (2.11) is by means of an illustration. Let me use one of my favorites, Hilbert's Hotel, a product of the mind of the great German mathematician, David Hilbert. Let us imagine a hotel with a finite number of rooms. Suppose, furthermore, that *all the rooms are full.* When a new guest arrives asking for a room, the proprietor apologizes, "Sorry, all the rooms are full." But now let us imagine a hotel with an infinite number of rooms and suppose once more that all the rooms are full. There is not a single vacant room throughout the entire infinite hotel. Now suppose a new guest shows up, asking for a room. "But of course!" says the proprietor, and he immediately shifts the person in room #1 into room #2, the person in room #2 into room #3, the person in room #3 into room #4, and so on, out to infinity. As a result of these room changes, room #1 now becomes vacant and the new guest gratefully checks in. But remember, before he arrived, all the rooms were full! Equally curious, according to the mathematicians, there are now no more persons in the hotel than there were before: the number is just infinite. But how can this be? The proprietor just added the new guest's name to the register and gave him his keys—how can there not be one more person in the hotel than before? But the situation becomes even stranger. For suppose an infinity of new guests show up the desk, asking for a room. "Of course, of course!" says the proprietor, and he proceeds to shift the person in room #1 into room #2, the person in room #2 into room #4, the person in room #3 into room #6, and so on out to infinity, always putting each former occupant into the room number twice his own. As a result, all the odd numbered rooms become vacant, and the infinity of new guests is easily accommodated. And yet, before they came, all the rooms were full! And again, strangely enough, the number of guests in the hotel is the same after the infinity of new guests check in as

before, even though there were as many new guests as old guests. In fact, the proprietor could repeat this process *infinitely many times*, and yet there would never be one single person more in the hotel than before.

But Hilbert's Hotel is even stranger than the German mathematician made it out to be. For suppose some of the guests start to check out. Suppose the guest in room #1 departs. Is there not now one less person in the hotel? Not according to the mathematicians—but just ask the woman who makes the beds! Suppose the guests in room numbers 1, 3, 5, etc., check out. In this case, an infinite number of people have left the hotel, but according to the mathematicians there are no less people in the hotel—but don't talk to that laundry woman! In fact, we could have every other guest check out of the hotel and repeat this process infinitely many times, and yet there would never be any less people in the hotel. . . . Can anyone sincerely believe that such a hotel could exist in reality? These sorts of absurdities illustrate the impossibility of the existence of an actually infinite number of things.

That takes us to (2.12). The truth of this premise seems fairly obvious. If the universe never began to exist, then prior to the present event there have existed an actually infinite number of previous events. Hence, a beginningless series of events in time entails the existence of an actually infinite number of things, namely, past events.

Given the truth of (2.11) and (2.12), the conclusion (2.13) logically follows. The series of past events must be finite and have a beginning. But since the universe is not distinct from the series of events, it follows that the universe began to exist.

At this point, we might find it profitable to consider several objections that might be raised against the argument. First, let us consider objections to (2.11). Wallace Matson objects that the premise must mean that an actually infinite number of things is *logically* impossible; but it is easy to show that such a collection is logically possible. For example, the series of negative numbers { . . . –3, –2, –1} is an actually infinite collection with no first member.[3] Matson's error here lies in thinking that (2.11) means to assert the *logical* impossibility of an actually infinite number of things. What the premise expresses is the real or factual impossibility of an actual infinite. . . .

The late J. L. Mackie also objected to (2.11), claiming that the absurdities are resolved by noting that for infinite groups the axiom "the whole is greater than its part" does not hold, as it does for finite groups.[4] Similarly, Quentin Smith comments that once we understand that an infinite set has a proper subset which has the same number of members as the set itself, the purportedly absurd situations become "perfectly believable."[5] But to my mind, it is precisely this feature of infinite set theory which, when translated into the realm of the real, yields results which are perfectly incredible, for example, Hilbert's Hotel. Moreover, not all the absurdities stem from infinite set theory's denial of Euclid's axiom: the absurdities illustrated by guests checking out of the hotel stem from the self-contradictory results when the inverse operations of subtraction or division are performed using transfinite numbers. Here the case against an actually infinite collection of things becomes decisive. . . .

With regard to (2.12), the most frequent objection is that the past ought to be regarded as a potential infinite only, not an actual infinite. This was Aquinas's position versus Bonaventure, and the contemporary philosopher Charles Hartshorne seems to side with Thomas on this issue.[6] Such a position is, however, untenable. The future is potentially infinite, since it does not exist; but the past is actual in a way the future is not, as evidenced by the fact that we have traces of the past in the present, but no traces of the future. Hence, if the series of past events never began to exist, there must have been an actually infinite number of past events.

The objections to either premise therefore seem to be less compelling than the premises themselves. Together they imply that the universe began to exist. Hence, I conclude that this argument furnishes good grounds for accepting the truth of premise (2) that the universe began to exist.

Second Supporting Argument

The second argument (2.2) for the beginning of the universe is based on the impossibility of forming an

actual infinite by successive addition. This argument is distinct from the first in that it does not deny the possibility of the existence of an actual infinite, but the possibility of its being *formed* by successive addition.

Premise (2.21) is the crucial step in the argument. One cannot form an actually infinite collection of things by successively adding one member after another. Since one can always add one more before arriving at infinity, it is impossible to reach actual infinity. Sometimes this is called the impossibility of "counting to infinity" or "traversing the infinite." It is important to understand that this impossibility has nothing to do with the amount of time available: it belongs to the nature of infinity that it cannot be so formed.

Now someone might say that while an infinite collection cannot be formed by beginning at a point and adding members, nevertheless an infinite collection could be formed by never beginning but ending at a point, that is to say, ending at a point after having added one member after another from eternity. But this method seems even more unbelievable than the first method. If one cannot count to infinity, how can one count down from infinity? If one cannot traverse the infinite by moving in one direction, how can one traverse it by simply moving in the opposite direction?

Indeed, the idea of a beginningless series ending in the present seems to be absurd. To give just one illustration: suppose we meet a man who claims to have been counting from eternity and is now finishing: . . . , –3, –2, –1, 0. We could ask, why did he not finish counting yesterday or the day before or the year before? By then, an infinite time had already elapsed, so that he should already have finished. Thus, at no point in the infinite past could we ever find the man finishing his countdown, for by that point he should already be done! In fact, no matter how far back into the past we go, we can never find the man counting at all, for at any point we reach, he will have already finished. But if at no point in the past do we find him counting, this contradicts the hypothesis that he has been counting from eternity. This illustrates the fact that the formation of an actual infinite by successive addition is equally impossible whether one proceeds to or from infinity. . . .

Given the truth of (2.21) and (2.22), the conclusion (2.23) logically follows. If the universe did not begin to exist a finite time ago, then the present moment could never arrive. But obviously, it has arrived. Therefore, we know that the universe is finite in the past and began to exist.

Again, it would be profitable to consider various objections that have been offered against this reasoning. Against (2.21), Mackie objects that the argument illicitly assumes an infinitely distant starting point in the past and then pronounces it impossible to travel from that point to today. But there would in an infinite past be no starting point, not even an infinitely distant one. Yet from any given point in the infinite past, there is only a finite distance to the present.[7] Now it seems to me that Mackie's allegation that the argument presupposes an infinitely distant starting point is entirely groundless. The beginningless character of the series only serves to accentuate the difficulty of its being formed by successive addition. The fact that there is *no beginning at all*, not even an infinitely distant one, makes the problem more, not less, nettlesome. And the point that from any moment in the infinite past there is only a finite temporal distance to the present may be dismissed as irrelevant. The question is not how any finite portion of the temporal series can be formed, but how the whole infinite series can be formed. If Mackie thinks that because every segment of the series can be formed by successive addition and therefore the whole series can be so formed, then he is simply committing the fallacy of composition. . . .

As for premise (2.22), many thinkers have objected that we need not regard the past as a beginningless infinite series with an end in the present. Popper, for example, admits that the *set* of all past events is actually infinite, but holds that the *series* of past events is potentially infinite. This may be seen by beginning in the present and numbering the events backwards, thus forming a potential infinite. Therefore, the problem of an actual infinite's being formed by successive addition does not arise. . . .[8] This objection, however, clearly confuses the *mental regress* of counting with the *real progress* of the temporal series of events itself. Numbering the series from the present backwards only shows that if there are an infinite number of past

events, then we can denumerate an infinite number of past events. But the problem is, how can this infinite collection of events come to be *formed* by successive addition? How we mentally conceive the series does not in any way affect the ontological character of the series itself as a series with no beginning but an end, or in other words, as an actual infinite completed by successive addition.

Once again, then, the objections to (2.21) and (2.22) seem less plausible than the premises themselves. Together they imply (2.23), or that the universe began to exist.

First Scientific Confirmation

These purely philosophical arguments for the beginning of the universe have received remarkable confirmation from discoveries in astronomy and astrophysics during this century. These confirmations might be summarized under two heads: the confirmation from the expansion of the universe and the confirmation from thermodynamic properties of the universe.

With regard to the first, Hubble's discovery in 1929 of the red-shift in the light from distant galaxies began a revolution in astronomy perhaps as significant as the Copernican revolution. Prior to this time, the universe as a whole was conceived to be static; but the startling conclusion to which Hubble was led was that the red-shift is due to the fact that the universe is in fact *expanding*. The staggering implication of this fact is that as one traces the expansion back in time, the universe becomes denser and denser until one reaches a point of infinite density from which the universe began to expand. The upshot of Hubble's discovery is that at some point in the finite past—probably around 15 billion years ago—the entire known universe was contracted down to a single mathematical point which marked the origin of the universe. That initial explosion has come to be known as the "Big Bang." Four of the world's most prominent astronomers described that event:

> The universe began from a state of infinite density. . . . Space and time were created in that event and so was all the matter in the universe. It is not meaningful to ask what happened before the Big Bang; it is like asking what is north of the North Pole. Similarly, it is not sensible to ask where the Big Bang took place. The point-universe was not an object isolated in space; it was the entire universe, and so the answer can only be that the Big Bang happened everywhere.[9]

This event that marked the beginning of the universe becomes all the more amazing when one reflects on the fact that a state of "infinite density" is synonymous to "nothing." There can be no object that possesses infinite density, for if it had any size at all it could still be even more dense. Therefore, as the Cambridge astronomer Fred Hoyle points out, the Big Bang theory requires the creation of matter from nothing. This is because as one goes back in time, one reaches a point at which, in Hoyle's words, the universe was "shrunk down to nothing at all."[10] Thus, what the Big Bang model of the universe seems to require is that the universe began to exist and was created out of nothing.

Some theorists have attempted to avoid the absolute beginning of the universe implied by the Big Bang theory by speculating that the universe may undergo an infinite series of expansions and contractions. There are, however, good grounds for doubting the adequacy of such an oscillating model of the universe: (i) The oscillating model appears to be physically impossible. For all the talk about such models, the fact seems to be that they are only theoretically, but not physically, possible. As the late Professor [Beatrice] Tinsley of Yale explains, in oscillating models, "even though the mathematics say that the universe oscillates, there is no known physics to reverse the collapse and bounce back to a new expansion. The physics seems to say that those models start from the Big Bang, expand, collapse, then end."[11] In order for the oscillating model to be correct, it would seem that the known laws of physics would have to be revised. (ii) The oscillating model seems to be observationally untenable. Two facts of observational astronomy appear to run contrary to the oscillating model. First, the observed homogeneity of matter distribution throughout the universe seems unaccountable on an oscillating model. During the contraction phase of such a model, black holes begin to gobble up surrounding matter, resulting in an inhomogeneous distribution of matter. But there is no known mechanism to "iron out" these inhomogeneities

during the ensuing expansion phase. Thus, the homogeneity of matter observed throughout the universe would remain unexplained. Second, the density of the universe appears to be insufficient for the re-contraction of the universe. For the oscillating model to be even possible, it is necessary that the universe be sufficiently dense such that gravity can overcome the force of the expansion and pull the universe back together again. However, according to the best estimates, if one takes into account both luminous matter and nonluminous matter (found in galactic halos) as well as any possible contribution of neutrino particles to total mass, the universe is still only about one-half of that needed for re-contraction.[12] Moreover, recent work on calculating the speed and deceleration of the expansion confirms that the universe is expanding at, so to speak, "escape velocity" and will not therefore re-contract. According to Sandage and Tammann, "Hence, we are forced to decide that . . . it seems inevitable that the Universe will expand forever"; they conclude, therefore, that "the Universe has happened only once."[13]

Second Scientific Confirmation

As if this were not enough, there is a second scientific confirmation of the beginning of the universe, based on the thermodynamic properties of various cosmological models. According to the second law of thermodynamics, processes taking place in a closed system always tend toward a state of equilibrium. Now our interest is in what implications this has when the law is applied to the universe as a whole. For the universe is a gigantic closed system, since it is everything there is and no energy is being fed into it from without. The second law seems to imply that, given enough time, the universe will reach a state of thermodynamic equilibrium, known as the "heat death" of the universe. This death may be hot or cold, depending on whether the universe will expand forever or eventually re-contract. On the one hand, if the density of the universe is great enough to overcome the force of the expansion, then the universe will re-contract into a hot fireball. As the universe contracts, the stars burn more rapidly until they finally explode or evaporate. As the universe grows denser, the black holes begin to gobble up everything around them and begin themselves to coalesce, until all the black holes finally coalesce into one gigantic black hole which is coextensive with the universe, from which it will never reemerge. On the other hand, if the density of the universe is insufficient to halt the expansion, as seems more likely, then the galaxies will turn all their gas into stars and the stars will burn out. At 10^{30} years, the universe will consist of 90 percent dead stars, 9 percent supermassive black holes, and 1 percent atomic matter. Elementary particle physics suggests that thereafter protons will decay into electrons and positrons, so that space will be filled with a rarefied gas so thin that the distance between an electron and a positron will be about the size of the present galaxy. At 10^{100} years some scientists believe that the black holes themselves will dissipate into radiation and elementary particles. Eventually, all the matter in the dark, cold, ever-expanding universe will be reduced to an ultrathin gas of elementary particles and radiation. Equilibrium will prevail throughout, and the entire universe will be in its final state, from which no change will occur.

Now the question which needs to be asked is this: if, given sufficient time, the universe will reach heat death, then why is it not now in a state of heat death if it has existed for infinite time? If the universe did not begin to exist, then it should now be in a state of equilibrium. Some theorists have suggested that the universe escapes final heat death by oscillating from eternity past to eternity future. But we have already seen that such a model seems to be physically and observationally untenable. But even if we waive those considerations and suppose that the universe does oscillate, the fact is that the thermodynamic properties of this model imply the very beginning of the universe which its proponents seek to avoid. For the thermodynamic properties of an oscillating model are such that the universe expands farther and farther with each successive cycle. Therefore, as one traces the expansions back in time, they grow smaller and smaller. As one scientific team explains, "The effect of entropy production will be to enlarge the cosmic scale, from cycle to cycle. . . . Thus, looking back in time, each cycle generated less entropy, had a smaller cycle time, and had a smaller cycle expansion

factor than the cycle that followed it."[14] Novikov and Zeldovich of the Institute of Applied Mathematics of the Russian Academy of Sciences therefore conclude, "The multicycle model has an infinite future, but only a finite past."[15] As another writer points out, the oscillating model of the universe thus still requires an origin of the universe prior to the smallest cycle.[16]

So whatever scenario one selects for the future of the universe, thermodynamics implies that the universe began to exist. According to physicist P. C. W. Davies, the universe must have been created a finite time ago and is in the process of winding down. Prior to the creation, the universe simply did not exist. Therefore, Davies concludes, even though we may not like it, we must conclude that the universe's energy was somehow simply "put in" at the creation as an initial condition.[17]

We therefore have both a philosophical argument and scientific confirmation for the beginning of the universe. On this basis, I think that we are amply justified in concluding the truth of premise (2) that the universe began to exist.

First Premise

Premise (1) strikes me as relatively noncontroversial. It is based on the metaphysical intuition that something cannot come out of nothing. Hence, any argument for the principle is apt to be less obvious than the principle itself. Even the great skeptic David Hume admitted that he never asserted so absurd a proposition as that something might come into existence without a cause; he only denied that one could *prove* the obviously true causal principle.[18] With regard to the universe, if originally there were absolutely *nothing*—no God, no space, no time—then how could the universe possibly come to exist? The truth of the principle *ex nihilo, nihil fit* (out of nothing, nothing comes) is so obvious that I think we are justified in foregoing an elaborate defense of the argument's first premise.

Nevertheless, some thinkers, exercised to avoid the theism implicit in this premise within the present context, have felt driven to deny its truth. In order to avoid its theistic implications, Davies presents a scenario which, he confesses, "should not be taken too seriously," but which seems to have a powerful attraction for Davies.[19] He has reference to a quantum theory of gravity according to which space-time itself could spring uncaused into being out of absolutely nothing. While admitting that there is "still no satisfactory theory of quantum gravity," such a theory "would allow spacetime to be created and destroyed spontaneously and uncaused in the same way that particles are created and destroyed spontaneously and uncaused. The theory would entail a certain mathematically determined probability that, for instance, a blob of space would appear where none existed before. Thus, spacetime could pop out of nothingness as the result of a causeless quantum transition."[20]

Now in fact particle pair production furnishes no analogy for this radical *ex nihilo* becoming, as Davies seems to imply. This quantum phenomenon, even if an exception to the principle that every event has a cause, provides no analogy to something's coming into being out of nothing. Although physicists speak of this as particle pair creation and annihilation, such terms are philosophically misleading, for all that actually occurs is conversion of energy into matter or vice versa. As Davies admits, "The processes described here do not represent the creation of matter out of nothing, but the conversion of preexisting energy into material form."[21] Hence, Davies greatly misleads his reader when he claims that "particles . . . can appear out of nowhere without specific causation" and again, "yet the world of quantum physics routinely produces something for nothing."[22] On the contrary, the world of quantum physics *never* produces something for nothing. . . .

Conclusion

Given the truth of premises (1) and (2), it logically follows that (3) the universe has a cause of its existence. In fact, I think that it can be plausibly argued that the cause of the universe must be a personal Creator. For how else could a temporal effect arise from an eternal cause? If the cause were simply a mechanically operating set of necessary and sufficient conditions existing from eternity, then why would not the effect also exist from eternity? . . . The only

way to have an eternal cause but a temporal effect would seem to be if the cause is a personal agent who freely chooses to create an effect in time. . . . Thus, we are brought not merely to the first cause of the universe, but to its personal Creator.

Summary and Conclusion

We have seen on the basis of both philosophical argument and scientific confirmation that it is plausible that the universe began to exist. Given the intuitively obvious principle that whatever begins to exist has a cause of its existence, we have been led to conclude that the universe has a cause of its existence. On the basis of our argument, this cause would have to be uncaused, eternal, changeless, timeless, and immaterial. Moreover, it would have to be a personal agent who freely elects to create an effect in time. Therefore, on the basis of the *kalam* cosmological argument, I conclude that it is rational to believe that God exists.

Notes

1. G. W. Leibniz, "The Principles of Nature and of Grace, Based on Reason," in *Leibniz Selections*, ed. Philip P. Wiener, *The Modern Student's Library* (New York: Scribners, 1951), p. 527.
2. See William Lane Craig, *The Cosmological Argument from Plato to Leibniz*, Library of Philosophy and Religion (London: Macmillan, 1980), pp. 48–58, 61–76, 98–104, 128–131.
3. Wallace Matson, *The Existence of God* (Ithaca, N.Y.: Cornell University Press, 1965), pp. 58–60.
4. J. L. Mackie, *The Miracle of Theism* (Oxford, England: Clarendon Press, 1982), p. 93.
5. Quentin Smith, "Infinity and the Past," *Philosophy of Science* 54 (1987): 69.
6. Charles Hartshorne, *Man's Vision of God and the Logic of Theism* (Chicago: Willett, Clark, & Co., 1941), p. 37.
7. Mackie, *Theism*, p. 93.
8. K. R. Popper, "On the Possibility of an Infinite Past: A Reply to Whitrow," *British Journal for the Philosophy of Science* 29 (1978): 47–48.
9. Richard J. Gott, et al., "Will the Universe Expand Forever?" *Scientific American* (March 1976), p. 65.
10. Fred Hoyle, *From Stonehenge to Modern Cosmology* (San Francisco, Calif: W. H. Freeman, 1972), p. 36.
11. Beatrice Tinsley, personal letter.
12. David N. Schramm and Gary Steigman, "Relic Neutrinos and the Density of the Universe," *Astrophysical Journal* 243 (1981): 1–7.
13. Alan Sandage and G. A. Tammann, "Steps toward the Hubble Constant: VII," *Astrophyscial Journal* 210 (1976): 23, 7; see also idem, "Steps toward the Hubble Constant: VIII," *Astrophysical Journal* 256 (1982): 339–345.
14. Duane Dicus, et al. "Effects of Proton Decay on the Cosmological Future." *Astrophysical Journal* 252 (1982): 1, 8.
15. I. D. Novikov and Ya. B. Zeldovich, "Physical Processes Near Cosmological Singularities," *Annual Review of Astronomy and Astrophysics* 11 (1973): 401–402.
16. John Gribbin, "Oscillating Universe Bounces Back," *Nature* 259 (1976): 16.
17. P. C. W. Davies, *The Physics of Time Asymmetry* (London: Surrey University Press, 1974), p. 104.
18. David Hume to John Stewart, February 1754, in *The Letters of David Hume*, ed. J. Y. T. Greig (Oxford, England: Clarendon Press, 1932), 1: 187.
19. Paul Davies, *God and the New Physics* (New York: Simon & Schuster, 1983), p. 214.
20. Ibid., p. 215.
21. Ibid., p. 31.
22. Ibid., pp. 215, 216.

Study Questions

1. What is the kalam cosmological argument? How does it differ from the version of the cosmological argument in the previous reading?
2. How does Craig argue against the possibility of an actual infinite?
3. What does the Big Bang theory have to do with showing that the universe had a beginning?
4. How does Craig argue that the first cause was a personal being?